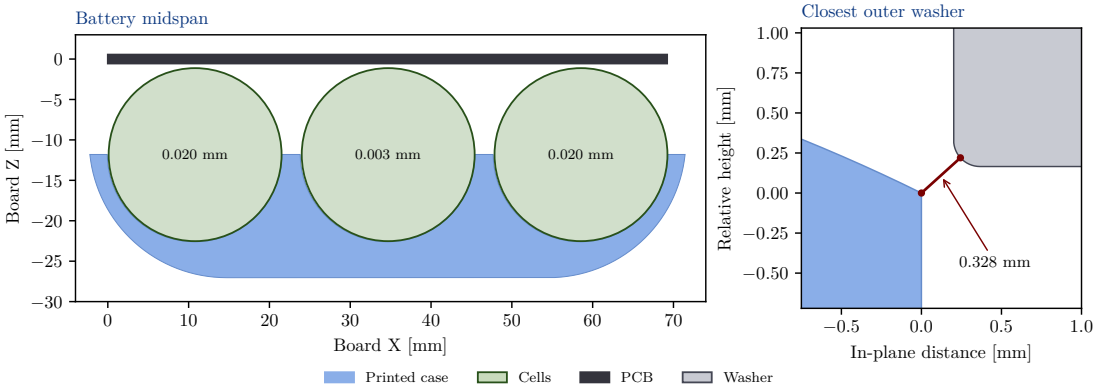
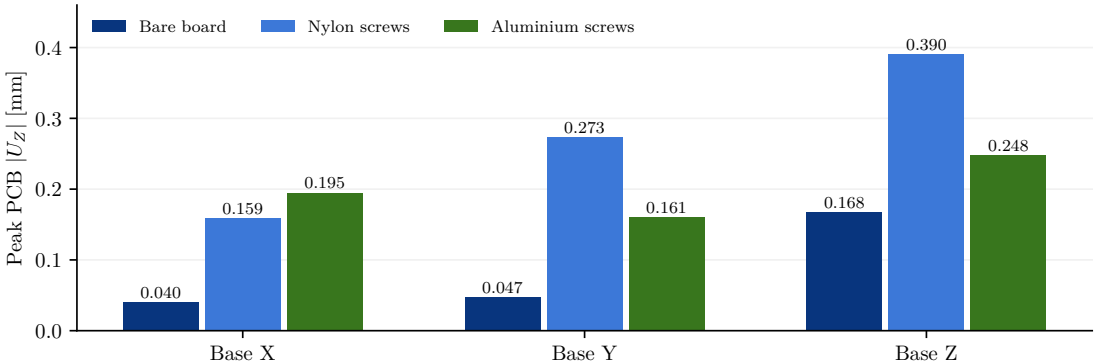
The three-pocket Balius case fits at 1.25 mm stand-off

Balius native case at 1.25 mm tab-to-board spacing



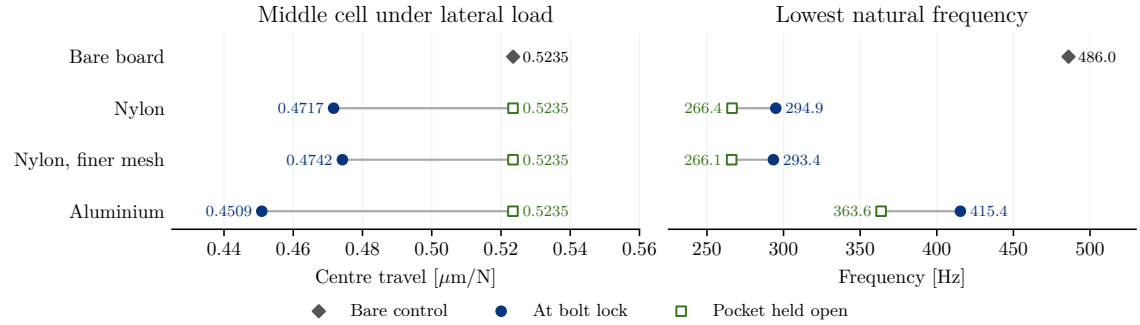
The dense PLA shell adds 73.7 g to the 251.8 g bare board. The middle-cell clearance is 0.003 mm and the closest washer clears by 0.328 mm. In the local pull model the left cell first reaches its pocket at about **1.14 N toward the middle cell** (0.020 mm) or **6.7 N away** (0.165 mm). The forces come from the bare strips up to contact.

The snug case increases board travel in the seeded clips



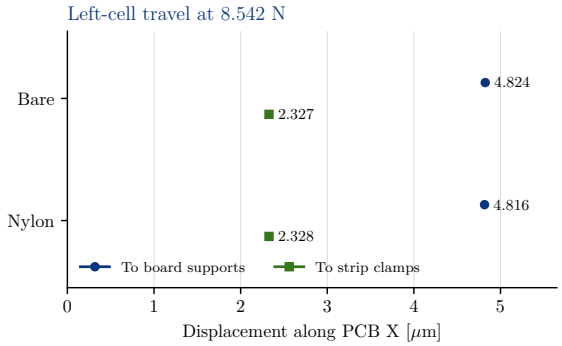
During the displayed base-Z interval, PCB travel normal to the board reached **0.168 mm bare**, **0.390 mm with nylon screws** and **0.248 mm with aluminium**. All 22 paired movie subjects are available.

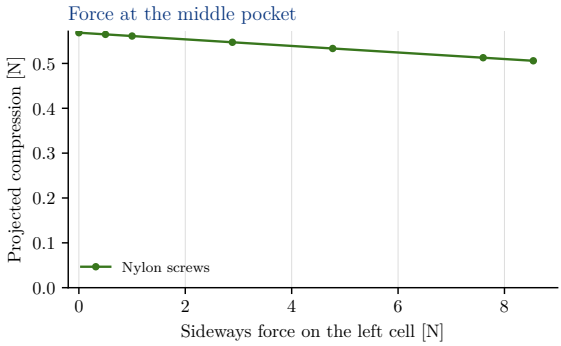
Middle-cell restraint depends on contact with the pocket



At bolt lock, the pocket took **10.4% of a small sideways load with nylon screws** and **14.5% with aluminium**. Holding the pocket open in the whole-board calculation brought cell travel within **0.011% of bare** with either mount. Loading the adjacent left cell opened the aluminium contact at **3.07 N**. No additional middle-pocket point touched through 8.54 N in the independent mesh calculation.

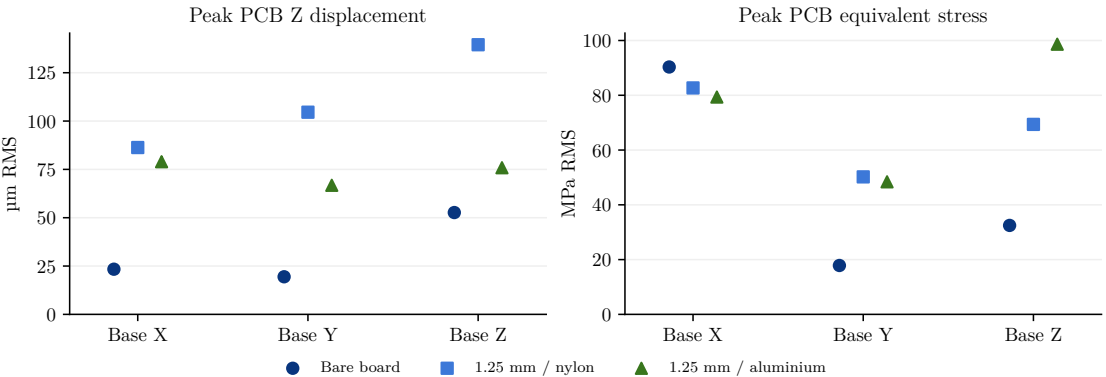
The local fixture and the complete board respond differently





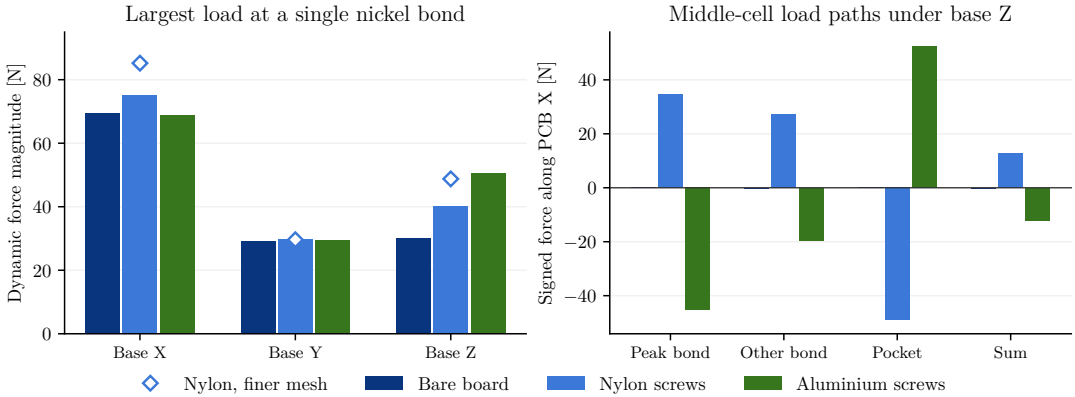
At **8.54 N**, the complete board gave **4.824 μm bare** and **4.816 μm with nylon screws**. The driven cell stayed clear of the pocket. In the local fixture it reached the case, reducing travel from **0.500 mm bare** to about **0.034 mm with nylon screws**.

Aluminium cuts board travel relative to nylon but raises base-Z stress



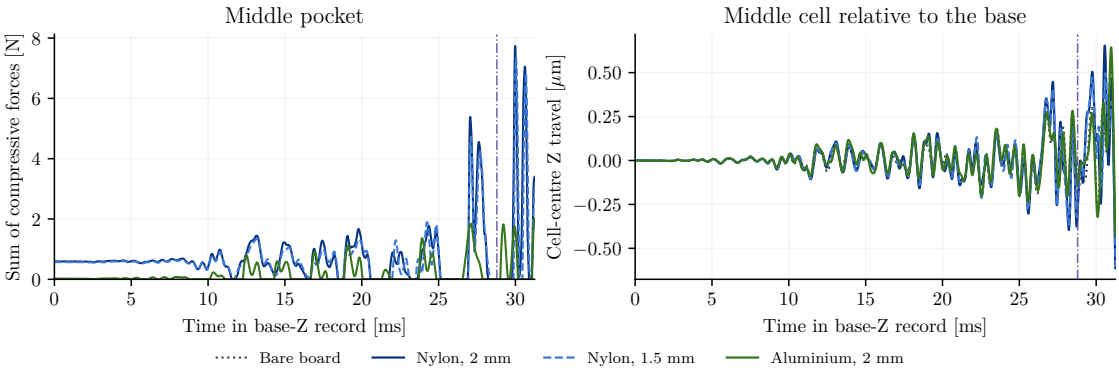
Under base Z, peak PCB stress reached **32.5 MPa RMS bare**, **69.4 with nylon** and **98.7 with aluminium**. Under base Y both mounts raised stress over nearly the entire board. These native spectra hold the installed contact state fixed.

Board-normal vibration loads the middle cell's nickel bond



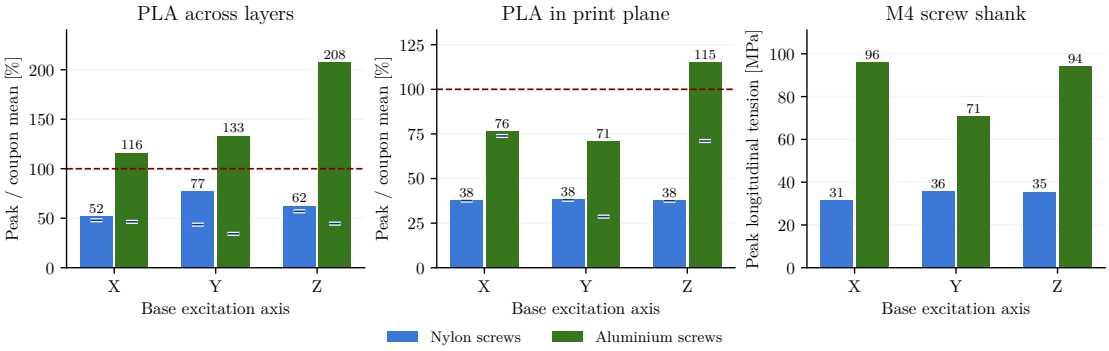
The largest base-Z bond load rose from **30.2 N bare** to **40.1 N with nylon screws** and **50.5 N with aluminium**. The finer nylon mesh gave **48.8 N**. All cased base-Z peaks occur at the middle cell. Bars use native histories at the installed contact state. The diamond uses independent integration on the finer native modes.

The middle pocket unloads and carries force again



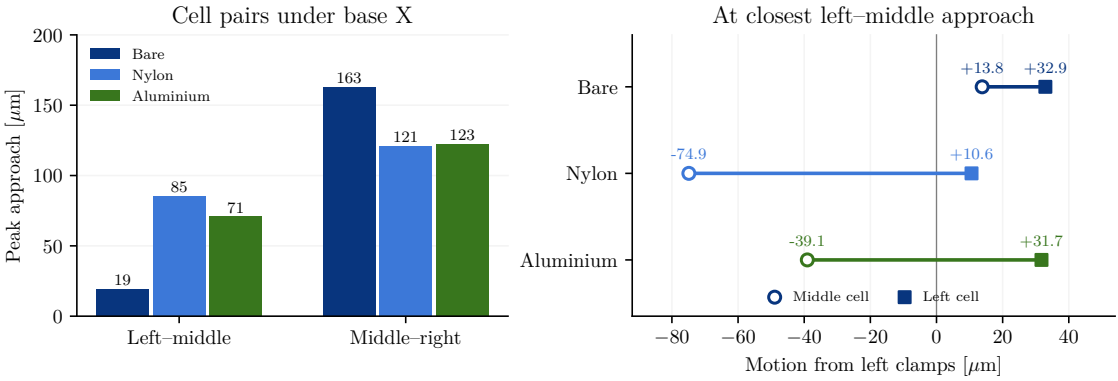
Under base Z, the first interval with all original pocket points unloaded began at **5.61 ms with aluminium** and **11.54 ms with nylon** (11.64 ms on finer nylon). At 28.79 ms the middle cell cleared the case by **3.90–4.01 μm with nylon** and **1.73–1.84 μm with aluminium** on the nominal meshes. New contacts could change this motion.

The printed tab exceeds its coupon reference with aluminium screws



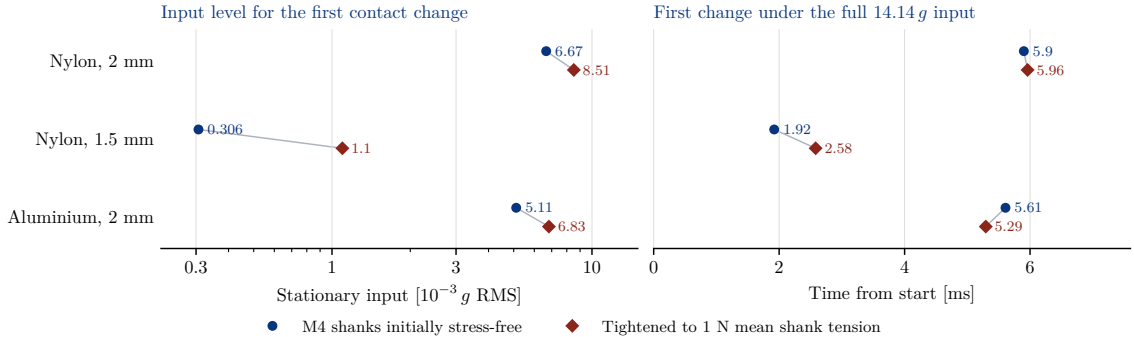
On the nominal mesh the tab reached **77% of the through-layer coupon mean with nylon** and **208% with aluminium**. Refining the nylon case shifted its maximum to **71.5% under base X**. The aluminium mesh check is queued. Horizontal marks exclude the 2 mm bearing band.

Under base X, the left and middle cells move closer together



The left cell's peak travel relative to its clamps fell by **29% with nylon** and **12% with aluminium**. Rotation accounted for **82%** of that travel in all three assemblies. With the middle cell free to move, peak approach rose from **19 μm bare** to **85 and 71 μm**. Contact stays fixed at bolt lock in these native histories.

M4 tension raises the vibration level for a pocket contact change



With **1 N mean M4 tension**, a new pocket point touched at about **0.0085** g **RMS with nylon** and **0.0068** g **with aluminium** on the 2 mm mesh. On finer nylon, a point that opened during tightening seated again at **0.0011** g. These are modeled contact changes under the board-normal waveform. Actual screw tension is unmeasured.